WASP-12b

R-0547

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_181020 · created 2026-07-22T06:27:51+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458411.89442 +/- 0.00055 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.094 +/- 0.011
Transit depth (Rp/Rs)^2	0.88 +/- 0.21 [%]
Orbital Inclination (inc)	83.5 +/- 2.27
Airmass coefficient 1 (a1)	1.1395 +/- 0.0098
Airmass coefficient 2 (a2)	-0.1186 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	0.85 %
Best Comparison Star	None
Optimal Aperture	4.66
Optimal Annulus	14.26
Transit Duration (day)	0.121 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.094 ± 0.011 vs 0.1178 (deviation 1.56σ)
Duration fit vs published	0.121 d vs 0.125 d (deviation 0.61σ)
Mid-time O–C	1.0 min
Depth SNR	6.2
Residual scatter	0.85 %

Light curve

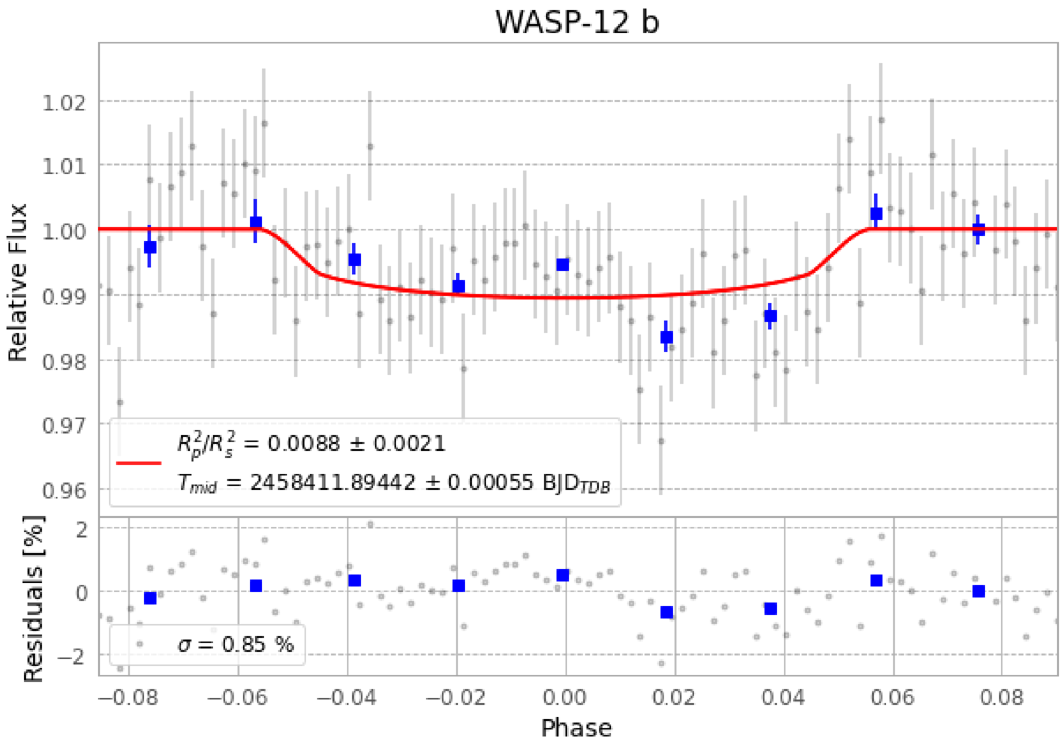

Figure 1 — FinalLightCurve_WASP-12 b_2018-10-20.png (SHA-256 606ab6cbbfd7d7c8…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 155], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e156036b9745bbe4…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

93 FITS frames (61 MB), `WASP-12181020070912.FITS` → `WASP-12181020114512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-181020.log (52f4f891b075…), FinalLightCurve_WASP-12 b_2018-10-20.png (606ab6cbbfd7…), FinalLightCurve_WASP-12 b_2018-10-20.csv (b7355af5c2d9…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)